

2.5.3

$$\dot{x} = rx + x^3$$

$$\int \frac{dx}{rx + x^3} = \int dt$$

$$-\frac{1}{2r} \ln \left[\frac{r+x^2}{x^2} \right] = t + c \quad \xRightarrow{t=0, x=x_0} c = -\frac{1}{2r} \ln \left[\frac{r+x_0^2}{x_0^2} \right]$$

$$\frac{r+x^2}{x^2} = e^{-2r(t+c)}$$

$$r = x^2 (e^{-2r(t+c)} - 1)$$

$$x = \pm \left(\frac{r}{e^{-2r(t+c)} - 1} \right)^{\frac{1}{2}}$$

if $e^{-2r(t+c)} \rightarrow 0$ Then $x \rightarrow \pm \infty$

so, when $t = -c$ (or, $t = \frac{1}{2r} \ln \left[\frac{r+x_0^2}{x_0^2} \right]$), $x \rightarrow \pm \infty$

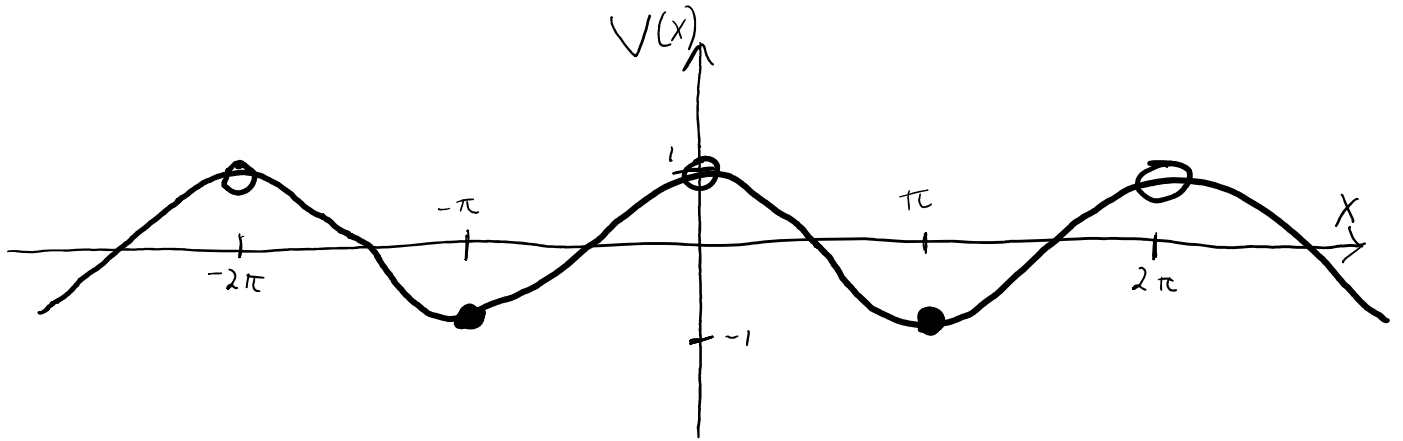
2.6.1

A harmonic oscillator is described by a 2nd order diff. eq.. So, it requires two numbers to specify the state. For instance, you need to specify where a pendulum is and which way it's swinging to know how the system will evolve. This means it actually has a 2-D state space.

2.7.3

$$\dot{X} = \sin[x]$$

$$\Rightarrow V(x) = \cos[x]$$

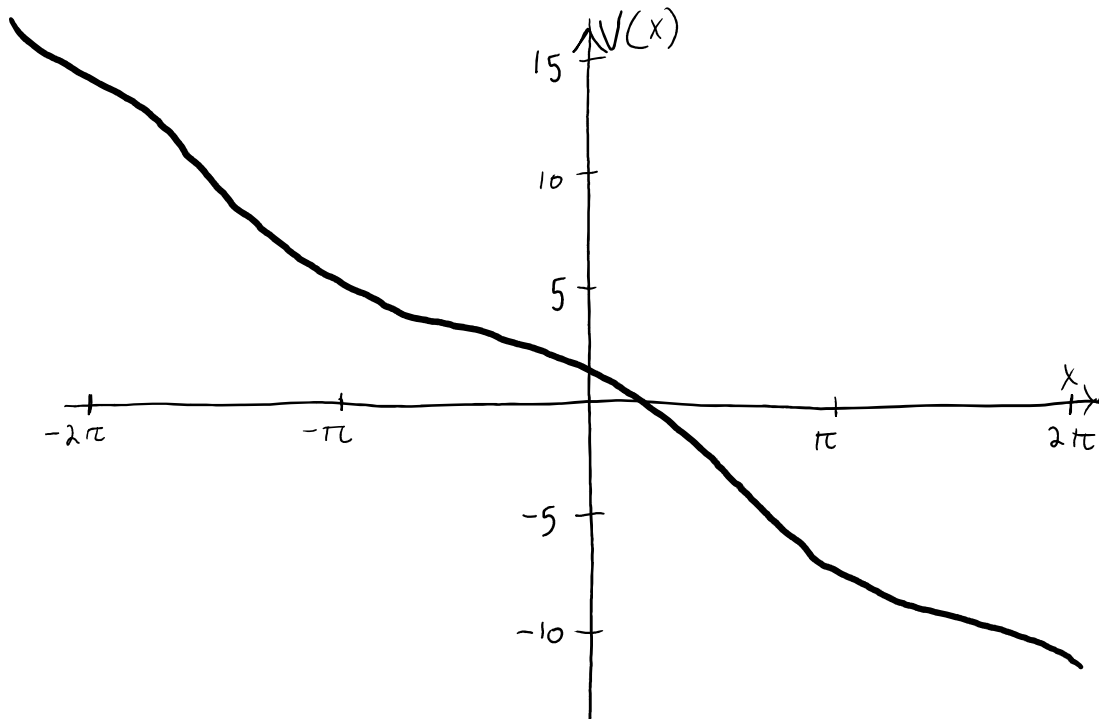


2.7.4

$$\dot{X} = \sin[x] + 2$$

$$\Rightarrow V(x) = \cos[x] - 2x$$

There are No equilibria.



Problem 2.8.3

Part A

$$\dot{x} = \frac{dx}{dt} = -x \iff \frac{dx}{x} = -dt \iff \log \frac{x}{x_0} = -t \iff x(t) = x_0 e^{-t} \underset{x_0=1}{\iff} x(t) = e^{-t}$$

So we find $x(1) = e^{-1}$

Part B

We can do $n = 0$, i.e. $\Delta t = 1$, by hand nicely so we do that and the rest in Matlab.

$$x_1 = x_0 + f(x_0)\Delta t = 1 + (-1)(1) = 0$$

So we have $x_1 \approx 0$ which isn't right.

$\Delta t = 10^{-1}, \hat{x}(1) = 0.34868$; $\Delta t = 10^{-2}, \hat{x}(1) = 0.36603$; $\Delta t = 10^{-3}, \hat{x}(1) = 0.3677$; $\Delta t = 10^{-4}, \hat{x}(1) = 0.36786$.

Part C

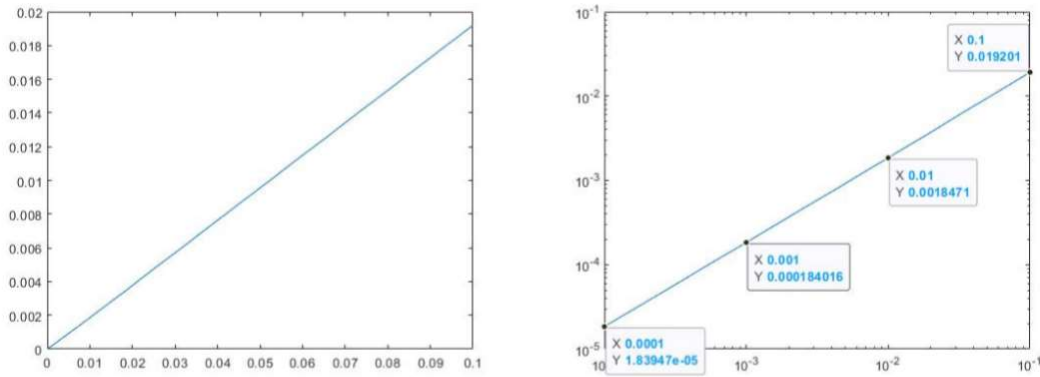


Figure 3: Δt vs E (Left) and $\log \Delta t$ vs $\log E$ (Right)

Both of these graphs have a near linear relationship between Δt and E then their logarithms as well. Hence the error is proportional to the step size.